

ON EXACTNESS OF SDP RELAXATION FOR THE MAXIMUM CUT PROBLEM

AVINASH BHARDWAJ*, HRITIZ GOGOI *, VISHNU NARAYANAN*, AND ABHISHEK PATHAPATI*

Abstract. Semidefinite programming provides a relaxation of the maximum cut problem on a graph. Given a graph $\mathcal{G}$ with rational weights, the decision problem of whether this relaxation is exact on $\mathcal{G}$ is known to be *NP*-complete since Delorme-Poljak (1993) [3], but its complexity was unresolved for unweighted graphs. In this work, we extend the *NP*-completeness result for unweighted graphs. We then investigate and present a few classes of graphs for which the semidefinite relaxation of the maximum cut problem is exact. For each class, we determine the optimal objective value by explicitly constructing a maximum cut and prove the uniqueness of the exact solution for two of these classes. Finally, we address two open problems posed by Mirka and Williamson (2024) [12] concerning the rank of the optimal solutions of an exact semidefinite relaxation, providing corresponding counterexamples.

1. Introduction. The Maximum-Cut (Max-Cut) problem asks the following : Given a graph $G = (V, E)$ with non-negative weights assigned to its edges, find an optimal partition of the vertex set V into two disjoint subsets such that the total weight of the edges crossing the partition (i.e., edges with endpoints in different subsets) is maximized. Over the years, this problem has generated immense interest. It is one of the first demonstrations of NP-hardness of computational problems [9] and the first optimization problem where semidefinite relaxation was used to propose an algorithm with an approximation guarantee [6]. Apart from its theoretical significance, the Max-Cut problem finds applications across various disciplines and industries. Prominent areas include statistical physics, VLSI design [1] and machine learning [17].

Goemans and Williamson [6] introduced a semidefinite programming (SDP) relaxation for the Max-Cut problem, which yields a polynomial-time approximation algorithm. Given an input graph, the relaxation produces a positive semidefinite matrix as the optimal solution to the primal problem. A feasible cut can then be obtained by applying a hyperplane rounding technique to this matrix. The authors demonstrated that the expected value of the resulting cut is at least a factor of $\alpha_{\text{GW}} = \frac{2}{\pi} \min_{0 \leq \theta \leq \pi} \frac{\theta}{1 - \cos \theta} \approx 0.87856$ times the value of the maximum cut. It was later shown that this approximation ratio is optimal under the assumption of the Unique Games Conjecture and the condition $P \neq NP$ [10]. The SDP relaxation is exact if and only if there exists a rank-1 positive semidefinite matrix as an optimal primal solution (see Subsection 2.3). One may, however, find it challenging to determine exactness solely from the obtained solution matrix. This limitation arises because the set of optimal solutions may include matrices of higher rank, and commercial solvers often return such high-rank solutions. Compounding this challenge, it is also possible to obtain sub-optimal feasible solutions even in cases where the relaxation is exact. Notably, Karloff [8] identified a class of graphs for which the semidefinite programming relaxation yields an exact solution; yet, the approximation ratio attained in these cases corresponds to the worst-case. Apart from the theoretical interests, these observations motivate the study of graph classes that guarantee the existence of a

*Industrial Engineering and Operations Research, Indian Institute of Technology Bombay, Mumbai, India 400076 (abhardwaj@iitb.ac.in, 22d1142@iitb.ac.in, vishnu@iitb.ac.in, apathapati@iitb.ac.in).

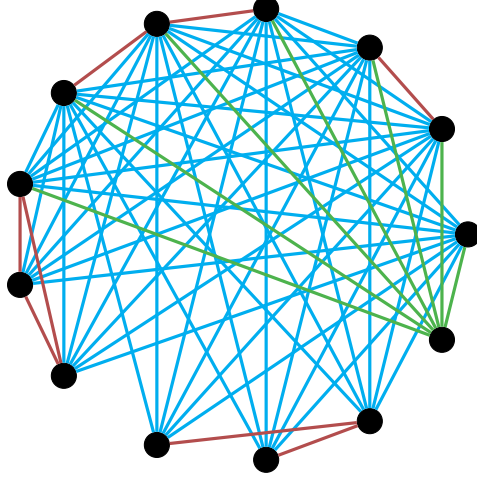


FIG. 1.1. Graph $\mathcal{G}$ with edge colors indicating weights: *Red* (1), *Blue* (2), and *Green* (6).

rank-1 optimal solution, thereby ensuring exactness of the relaxation. In this manuscript, we investigate and present a few such graph classes. For each of these classes, we determine the optimal objective value by explicitly constructing a maximum cut. Additionally, for two of these classes, we prove that the exact solution obtained via the relaxation is unique.

Example 1. Consider the graph $\mathcal{G}$ in Figure 1.1 as an example. Can we examine the structure of $\mathcal{G}$ and decompose it into simpler components to reveal whether the GW-relaxation of the Max-Cut problem is exact? It is thus, we are motivated to explore the following: Given a graph, can its exactness be inferred from its structure? Alternatively phrased, are there graph structures from which we can conclude if the graph is exact? In this manuscript, we develop and contribute to the set of tools and techniques to address these questions.

To place our contributions in context, we briefly revisit foundational works in the Max-Cut literature, particularly those of Svatopluk (Svata) Poljak and collaborators. Mohar and Poljak [13] provided an early result, deriving an eigenvalue-based upper bound on the optimal Max-Cut value, which served as a foundation for subsequent relaxation techniques. Delorme and Poljak [4] improved this upper bound by introducing a correcting vector. Delorme and Poljak [3, 4, 5] analyzed the performance of this improved bound (henceforth referred to as the *Delorme-Poljak bound*) on various graphs and graph classes. These studies also identified several classes of graphs where the Delorme-Poljak bound is tight, which we summarize in Subsection 2.5. They further proved that the Exact Weighted Graph Problem—whether the Delorme-Poljak bound coincides with the actual value of Max-Cut given a graph with rational weights—is *NP*-complete [3]. Building on these developments, Poljak and Rendl [14] proposed a semidefinite formulation for the eigenvalue approximation problem. Laurent and Poljak [11] investigated the feasible region of the semidefinite relaxation of the Max-Cut problem, the elliptope, and provided a complete characterization of objective functions of the form $C = bb^\top$, for some vector b , for which the relaxation is exact. Goemans and Williamson [6] developed the semidefinite relax-

ation of the quadratic integer formulation of Max-Cut (now known as GW-relaxation) and demonstrated its equivalence to the formulation obtained from Delorme-Poljak eigenvalue relaxation. Due to this equivalence, tightness of Delorme-Poljak bound implies exactness of GW-relaxation. Goemans and Williamson [6] also showed that the approximation ratio of the algorithm to be at least α_{GW} . Karloff [8] showed that the approximation ratio of GW-relaxation is exactly α_{GW} . Sojoudi and Lavaei [15] investigated the exactness of semidefinite relaxation for general quadratically constrained quadratic programs (QCQPs), of which the Max-Cut problem is a special case. More recently, Hong et al. [7] characterized conditions under which GW-relaxation is exact on weighted cycle graphs. They also derived bounds on the rank of the optimal solution for the vertex-sum of two graphs. Mirka and Williamson [12] investigated the uniqueness of the optimal rank-1 solution for the GW-relaxation. We summarize the graph classes identified in the literature for which the Delorme-Poljak eigenvalue bound is tight, or equivalently, the GW-relaxation is exact, in [Subsection 2.5](#).

This work is organized in following sections. [Section 2](#) provides necessary definitions, discusses some known results and highlights the notation used in this manuscript. [Section 3](#) outlines our main results and ends with examining the example graph $\mathcal{G}$. In [Section 4](#), we address two open problems posed by Mirka and Williamson (2024) concerning the rank of optimal solutions in exact GW-relaxation. In [Section 5](#), we provide the necessary lemmas and proofs for the main results. [Section 6](#) concludes this manuscript with a summary of our contributions and potential directions for future work.

2. Preliminaries. Given $A \in \mathbb{R}^{n \times n}$, we use $A \succeq \mathbf{0}$ ($A \succ \mathbf{0}$) to denote positive semidefiniteness (positive definiteness) of A . Through $\mathcal{N}(A)$, $\text{range}(A)$, $\text{tr}(A)$, $\lambda_{\max}(A)$ and $\lambda_{\min}(A)$, we denote the nullspace, range, trace, maximum eigenvalue and minimum eigenvalue, respectively for a matrix A . Kronecker product of two matrices A and B is denoted by $A \otimes B$. $\mathbf{I}_n$ represents $n \times n$ identity matrix, $\mathbf{1}_m$ represents an m -dimensional all-ones vector and $\mathbf{1}_{m \times n}$ represents an $m \times n$ all-ones matrix. Given a graph $G(V, E)$ and $S \subset V$, $\delta(S)$ denotes the cut induced by the partition $(S, V \setminus S)$. Value of this cut is denoted by $|\delta(S)|$ and the value of max-cut is denoted by $\text{MAXCUT}(G)$. A vector $y_S \in \mathbb{R}^n$ is called indicator vector of $S \subset V$ if $y_S(j) = +1$ for $j \in S$ and $y_S(j) = -1$ for $j \in V \setminus S$. By $\mathbb{Z}^+$, we denote the set of positive integers. For a matrix $B \in \mathbb{R}^{p \times q}$ and positive integers $m \leq p$, $n \leq q$, we denote by $B \upharpoonright_{m \times n}$ the $m \times n$ submatrix obtained by restricting B to its first m rows and first n columns. We say a graph $G(V, E)$ is *unweighted* if $w_{ij} = 1$ for all $(i, j) \in E$. Unless stated otherwise, all the graphs in this manuscript are simple and unweighted.

2.1. Laplacian of a Graph. Laplacian of a Graph G (denoted by $\mathcal{L}_G$ in this text) is a matrix with elements x_{ij} defined as: $x_{ij} = -w_{ij}$ if $i \neq j$ and $x_{ii} = \sum_{i \neq j} w_{ij}$. Likewise, for unweighted graphs, $x_{ij} = -1$ (for each edge between nodes i and j) and x_{ii} = degree of node i . Following are some useful properties of Laplacian of a graph:

1. For any indicator vector y_S of $S \subset V$, $y_S^\top \mathcal{L}_G y_S = 4 \times |\delta(S)|$.
2. For unweighted graph G , $\lambda_{\min}(\mathcal{L}_G) = 0$, $\lambda_{\max}(\mathcal{L}_G) \leq n$, where $|V| = n$ and $\lambda_{\max}(\mathcal{L}_G) \leq \max(\text{degree } v + \text{degree } u)$ where maximum is taken over all edges (u, v) . This bound is tight if and only if G is bipartite regular.

2.2. Eigenvalue bound on the Max Cut. Delorme and Poljak [4] formulated the following upper bound on Max-Cut.

$$\begin{aligned} \text{minimize } z_{\text{MAXCUTEIG}} &= \frac{n}{4} \lambda_{\max}(\mathcal{L}_G + \text{diag}(\mathbf{u})) \\ (\text{MAXCUT EIGEN}) \quad \text{subject to} \quad &\sum_i u_i = 0 \end{aligned}$$

Given a graph G and weights on its edges w_{ij} , optimal $z_{\text{MAXCUTEIG}}^* = \phi(G)$ has the following properties:

1. **Upper bound:** $\phi(G) \geq \text{MAXCUT}(G)$.
2. **Amalgamation:** If G_{12} is an amalgamation of G_1 and G_2 , then $\phi(G_1) + \phi(G_2) \geq \phi(G_{12})$. This bound is tight if G_1 and G_2 has atmost one common vertex.
3. **Scaling:** If each weight w_{ij} of G is multiplied by some positive real k , then $\phi(G)$ is also multiplied by k .
4. **Monotonicity:** If G and G' have the same vertices and the weight functions w and w' satisfy $w_{ij} \leq w'_{ij}$ for all pairs of vertices, then $\phi(G) \leq \phi(G')$.
5. **Graph Split:** Let $p_i \in \mathbb{Z}^+$ for $1 \leq i \leq n$. We construct a graph $\tilde{G}$ from G on $\sum_i^n p_i$ vertices as follows. Split vertex i of G into p_i new vertices. For each edge ij of G , insert in $\tilde{G}$ the complete bipartite graph connecting the p_i copies of vertex i to the p_j copies of vertex j . The weights of these $p_i p_j$ edges are defined as $w_{ij}/(p_i p_j)$. Then, $\phi(\tilde{G}) = \phi(G)$ for every splitting $\tilde{G}$ of G .

(For proofs, See [4], [3]).

2.3. Semidefinite Relaxation of Max Cut. The Max-Cut problem can be formulated as a quadratic integer programming problem:

$$\begin{aligned} \text{maximize } z_{\text{MAXCUTQIP}} &= \frac{1}{2} \sum_{i < j} w_{ij} (1 - y_i y_j) \\ (\text{MAXCUT QIP}) \quad \text{subject to} \quad &y_i \in \{-1, 1\}, \forall i \in V \end{aligned}$$

Since the expression $\frac{1}{2}(1 - y_i y_j)$ is non-zero if and only if $y_i \neq y_j$, the summation effectively includes only the weights of those edges whose incident vertices are assigned different values. Consequently, the optimization problem reduces to determining an assignment of the values 1 and -1 to the vertices that maximizes the objective. This is a quadratic integer optimization problem, which is known to be *NP*-Hard. In GW-relaxation, the binary assignment of each vertex is extended to a continuous setting by associating each vertex with a unit vector on the sphere in $\mathbb{R}^m$, for some $m \leq n$, where n is the number of vertices in the graph. The discrete multiplicative term $y_i y_j$ is accordingly generalized to the inner product $\langle v_i, v_j \rangle$ between the vectors assigned to vertices i and j . This relaxation yields a semidefinite program, which can be solved in polynomial time using standard convex optimization techniques.

Thus, the program MAXCUT QIP can be reformulated as a semidefinite program using Laplacian $\mathcal{L}_G$:

$$\begin{aligned} \text{maximize } z_{\text{MAXCUT}} &= \frac{1}{4} \langle \mathcal{L}_G, \mathbf{X} \rangle \\ (\text{MAXCUT SDP}) \quad \text{subject to} \quad &X_{ii} = 1, \quad \forall i \in V \\ &\mathbf{X} \succeq \mathbf{0} \end{aligned}$$

where $\mathbf{X} = (v_{ij})$ matrix is formed by taking the inner product of v_i and v_j for the $(i, j)^{th}$ element. It is straightforward to verify that, if we impose $\mathbf{X}$ to be a rank one matrix constraint in MAXCUT SDP, the solution is exact. Nonetheless, it remains relaxed as it is a non-convex constraint. MAXCUT SDP has the following dual:

$$\begin{aligned} \text{minimize } z_{\text{MAXCUTDUAL}} &= \text{tr}(\mathbf{Y}) \\ (\text{MAXCUT DUAL SDP}) \quad \text{subject to } \mathbf{S} &= \mathbf{Y} - \frac{1}{4}\mathcal{L}_G \\ \mathbf{S} &\succeq \mathbf{0} \end{aligned}$$

Poljak and Rendl [14] and Goemans and Williamson [6] showed the equivalence of MAXCUT SDP and MAXCUT EIGEN. Due to this equivalence, the optimal value of MAXCUT SDP (z_{MAXCUT}^*) inherits the properties of MAXCUT EIGEN's optimal value ($z_{\text{MAXCUTEIG}}^*$). We say a graph G is *exact* if an optimal rank-1 MAXCUT SDP solution exists for the graph or equivalently, if the Delorme-Poljak eigenvalue bound is tight. MAXCUT SDP obey principle of strong duality [16].

2.4. Definitions. Following are some preliminary objects we use in this manuscript:

DEFINITION 2.1 (Complete k-partite graph). *A complete k-partite graph is a graph whose vertices can be partitioned into k distinct independent sets with an edge between every pair of vertices from different independent sets.*

We denote $K(m_1, m_2, \dots, m_k)$ for a complete k-partite graph, where $m_1, m_2, \dots, m_k$ are the cardinalities of its independent sets and $m_1 \leq m_2 \leq \dots \leq m_k$. Complete bipartite graphs are complete k-partite graphs with $k = 2$.

DEFINITION 2.2 (Graph Join). *Graph Join of two disjoint graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$, denoted as $G_1 + G_2$, is the graph:*

$$G_1 + G_2 := (V_1 \cup V_2, E_1 \cup E_2 \cup \{(u, v) \mid u \in V_1, v \in V_2\})$$

DEFINITION 2.3 (Lexicographic Product of Graphs). *The lexicographic product of two graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$, is a graph $G_1 \circ G_2$ where:*

- The vertex set is $V(G_1 \circ G_2) = V_1 \times V_2$, the Cartesian product of V_1 and V_2 .
- Two vertices (u, v) and (u', v') in $V(G_1 \circ G_2)$ are adjacent if and only if:
 - u is adjacent to u' in G_1 , or
 - $u = u'$ and v is adjacent to v' in G_2 .

2.5. Known exact graphs. We present below a concise overview of graph classes previously identified in the literature for which the MAXCUT SDP relaxation is exact:

By Delorme and Poljak ([3, 4, 5]):

1. Cartesian product of two exact graphs.
2. Categorical product of complete graphs $K_m \otimes K_n$ if $n \leq m$, m is even.
3. Line graph of bipartite (r, s) -semiregular graphs where both r and s are even.
4. Line graph of complete graphs of the form K_{4r+1} .
5. Complement of line graph of bipartite graph $G(m, n)$ is exact if $m \leq n$ and m is even.
6. If all bi-connected components of a graph G are exact, then G is exact.

7. If G and G' are exact and max-cut of G and G' , respectively can be realized by a partition which coincide on $V \cap V'$. Then, amalgam G'' of G and G' is also exact.
8. Cayley graphs are exact if the weights are attached to the group $\mathcal{G} = \mathbb{Z}_2^m$, i.e. $w_{uv} = w_{u+v}$.

By Laurent and Poljak [11]:

9. Let $m_1 \leq m_2 \leq \dots \leq m_n \in \mathbb{Z}^+$. Let K_n be a complete graph with weights on each edge (i, j) is given by the corresponding pair $m_i m_j$. Then, K_n is exact if either of the following holds:
 - (a) There exists an index j such that $\sum_{1 \leq i < j} m_i = \sum_{j \leq i \leq n} m_i$. Equivalently, complete k -partite graph $K(m_1 \leq m_2 \leq \dots \leq m_k)$ is exact if there exists an index j such that $\sum_{1 \leq i < j} m_i = \sum_{j \leq i \leq k} m_i$.
 - (b) There exists an index k such that $m_k > \sum_{i \neq k} m_i$. Equivalently, complete k -partite graph $K(m_1 \leq m_2 \leq \dots \leq m_k)$ is exact if $m_k > \sum_{i \neq k} m_i$.

Remark 2.4. While Laurent and Poljak [11, Theorem 3.3] present their result in terms of the objective function, it can be verified that the weighted complete graphs in (9a) and (9b) correspond to the balanced and unbalanced cases of the theorem, respectively. Moreover, note that any complete k -partite graph can be viewed as the split graph of a corresponding weighted complete graph. See Lemma 3.1 for a discussion of the exactness equivalence between a graph G and its split graph $\tilde{G}$.

By Karloff [8]:

10. Let $J(m, t, b)$ be a graph such that:
 - (a) Vertex set is the set of all $\binom{m}{t}$ t -element subsets of $\{1, 2, \dots, m\}$.
 - (b) Two subsets S and T are adjacent if and only if $|S \cap T| = b$ and $S \neq T$.
 Then, $J(m, t, b)$ is exact if $t = \frac{m}{2}$, $b \leq \frac{m}{12}$ and m is even.

By Hong et al. [7]:

11. Weighted cycle graphs C_n are exact if and only if atleast one of the following is true:
 - (a) There are an even number of positively weighted edges in C_n .
 - (b) There exists an m such that $\frac{1}{|w_{m, m+1}|} \geq \sum_{i \neq m} \frac{1}{|w_{i, i+1}|}$.

In addition, Hong et al. [7] proved that the optimal solution of GW-relaxation for weighted cycle graphs given the aforementioned condition is also unique. Mirka and Williamson [12] proved the uniqueness of optimal solution of GW-relaxation for non-negatively weighted bipartite graphs.

2.6. Some useful results. We now state few results that are invoked in the proofs of the main results:

1. If A, B are symmetric matrices, $A - B \succeq 0$ if $\lambda_{\min}(A) - \lambda_{\max}(B) \geq 0$.
2. Consider a symmetric Matrix M of the form $M = \begin{bmatrix} A & B \\ B^\top & C \end{bmatrix}$, where A, B and C are symmetric matrices of appropriate dimensions. The following results hold:
 - (a) If $C \succ 0$, $M \succeq \mathbf{0}$ iff $A - BC^{-1}B^\top \succeq \mathbf{0}$.
 - (b) If $A \succeq 0$, $M \succeq \mathbf{0}$ iff $\text{range}(B) \subseteq \text{range}(A)$ and $C - B^\top A^+ B \succeq \mathbf{0}$, where

- A^+ indicates the Moore-Penrose pseudo-inverse of matrix A .
- (c) If $C \succ 0$, $\text{rank}(M) = \text{rank}(C) + \text{rank}(A - BC^{-1}B^\top)$ (Guttman's rank additivity).
3. For any positive semidefinite matrices A, B : $\lambda_{\max}(A \otimes B) = \lambda_{\max}(A) \times \lambda_{\max}(B)$ and $\lambda_{\min}(A \otimes B) = \lambda_{\min}(A) \times \lambda_{\min}(B)$.

3. Main Results. As mentioned in the introduction, Delorme and Poljak (1993) [3] established that the decision problem whether the Delorme-Poljak bound is tight (or equivalently the GW-relaxation is exact) for a graph with rational weights is NP -complete. However, the complexity of recognizing exact *unweighted* graphs remained open. To begin with, we claim that the problem is also NP -complete for unweighted graphs. The reduction method proceeds along the lines of the approach developed by Delorme and Poljak for graphs with rational weights. Their proof technique does not directly apply to the unweighted case, as it corresponds to the set $S = \{1, 1, \dots, 1\}$, for which the exact sum problem is trivial. In order to extend their framework, we incorporate the following lemma:

LEMMA 3.1. *Let $\tilde{G}$ be any split graph of G . Then, G is exact if and only if $\tilde{G}$ is exact.*

Proof. Let G be exact. A cut in G naturally induces a cut in $\tilde{G}$ with the same value, and hence $\text{MAXCUT}(\tilde{G}) \geq \text{MAXCUT}(G)$. Since the SDP optimum always serves as an upper bound on the maximum cut, it follows that $\phi(\tilde{G}) \geq \text{MAXCUT}(\tilde{G})$. Moreover, as $\tilde{G}$ is the split graph of G , we have $\phi(G) = \phi(\tilde{G})$. Combining these observations with exactness of G , we conclude that $\text{MAXCUT}(G) = \phi(G) = \phi(\tilde{G}) = \text{MAXCUT}(\tilde{G})$. Thus, $\tilde{G}$ is also exact.

Conversely, we claim that there always exists a partition corresponding to the maximum cut of $\tilde{G}$ in which all vertices of $\tilde{G}$ originating from the same vertex of G lie on the same side. Assume, for contradiction, that some vertex $v_j \in G$ is split across both sides in a maximum cut $(S, V \setminus S)$ of $\tilde{G}$, with its copies distributed across the two parts. Let the total weight of cut edges incident to each of the copies of v_j in S be m , and the total weight incident to each of those in $V \setminus S$ be n . If $m < n$ (resp. $m > n$), moving the copies from S to $V \setminus S$ (resp. $V \setminus S$ to S) strictly increases the cut value contradicting the maximality of the original cut, while if $m = n$, moving all copies to one side preserves the cut value. Therefore, there must exist a maximum cut of $\tilde{G}$ which groups the copies of each vertex together. Projecting this cut to G then yields a cut of the same weight, and the rest of the argument follows as in the other direction. $\square$

We are now ready to prove our claim. For the proof, we take an instance of the Exact Sum problem, which is known to be NP -complete, and reduce it to an instance of the recognition problem for exact graphs on an unweighted graph.

DEFINITION 3.2 (Exact Sum Problem). *Given positive integers $a_1, a_2, \dots, a_n$, does there exist a subset $S \subseteq \{1, 2, \dots, n\}$ such that $\sum_{i \in S} a_i = \sum_{i \notin S} a_i$?*

THEOREM 3.3. *Given any unweighted graph G , the recognition of exact graphs problem: “is MAXCUT SDP exact on G ?” is NP complete.*

Proof. Consider an instance of the Exact Sum problem. Assume that there is no index i for which $a_i > \sum_{j \neq i} a_j$, since in such a case, the answer to the Exact Sum problem is trivially negative.

Let $\tilde{H} = K(a_1, a_2, \dots, a_n)$ be an unweighted complete k -partite graph. We argue that $\tilde{H}$ is exact if and only if there exists a solution to the Exact Sum problem.

Note that every complete k -partite graph $\tilde{H}$ corresponds to a split graph of a weighted complete graph H with edge weights $w_{ij} = a_i a_j$. By [Lemma 3.1](#), $\tilde{H}$ is exact if and only if H is exact. As discussed in [Subsection 2.5](#), H is exact if and only if there exists a subset $S \subseteq \{1, 2, \dots, n\}$ such that $\sum_{i \in S} a_i = \sum_{i \notin S} a_i$.

Finally, showing that the recognition problem for unweighted exact graphs is in NP follows from using a reference rank-1 optimal matrix and a corresponding dual matrix as certificates, by verifying their feasibility and comparing the objective values. $\square$

This result contextualizes the following results in which we identify a few families of unweighted exact graphs, that is, graphs for which the optimal MAXCUT SDP solution coincides with the maximum cut. We defer the proofs to [Section 5](#).

3.1. Exact Graphs: Some Classes and Examples. We begin by observing that for a balanced complete bipartite graph $K(n, n)$, adding edges within the two independent sets does not affect the optimal solution for MAXCUT SDP; and the graph remains exact regardless of these additional edges.

PROPOSITION 3.4. *Let $G_K(V_K, E_K)$ be a complete bipartite graph $K(n, n)$, where $V_K = V_A \cup V_B$ is the disjoint union of independent sets V_A and V_B . Let $G_A(V_A, E_A)$ and $G_B(V_B, E_B)$ be any graphs with vertices in V_A and V_B , respectively. Let $G' = G_K \cup G_A \cup G_B$. Then G' is exact.*

It can be shown that the following statements are equivalent to [Proposition 3.4](#):

- Graphs with balanced complete bipartite spanning subgraph are exact.
- The graph join of any two graphs of identical order is exact.

COROLLARY 3.5. *The graph join of any graph with itself is exact.*

Observe that [Proposition 3.4](#) also explains 9(a) in [Subsection 2.5](#). Building on this, we next show that the graph join of two graphs can be exact even when their orders differ, provided that each graph contains a balanced complete bipartite spanning subgraph.

PROPOSITION 3.6. *Let $G' = G_K \cup G_A \cup G_B$ be a graph with complete bipartite spanning subgraph G_K with independent sets V_A and V_B . Let $G_A(V_A, E_A)$ and $G_B(V_B, E_B)$ be graphs with vertices in V_A and V_B , respectively. Then G' is exact if G_A and G_B both have balanced complete bipartite spanning subgraphs of order $|V_A|$ and $|V_B|$, respectively.*

Unlike [Proposition 3.4](#), where the structure of subgraphs G_A and G_B was inconsequential for their join to be an exact graph, [Proposition 3.6](#) requires both G_A and G_B to have a prescribed structure. Consequently, exactness is achieved if G_A and G_B belong to any of the following graph classes. This list is non-exhaustive, and both graphs may originate from the same class.

COROLLARY 3.7. *The graph join of any two graphs belonging to the following classes is exact:*

- Even cycles;
- Even complete graphs;
- Diamond graph;
- Circulant graph $C_{n,3}$ if n is even;

- Complete k -partite graph $K(m_1, \dots, m_{k-1}, m_k)$ if there exists a j such that $\sum_{1 \leq i < j} m_i = \sum_{j \leq i \leq k} m_i$.

Remark 3.8. It should be noted that the graph join of exact graphs (including bipartite graphs) is not necessarily exact. The graph join of the following pairs are few such examples:

- Complete graph on 4 vertices and path graph on 3 vertices: $K_4 + P_3$.
- Cycle graph on 4 vertices and path graph on 3 vertices: $C_4 + P_3$.
- Path graph on 2 and 3 vertices: $P_2 + P_3$.

Continuing the identification of exact graph joins, we observe that the graph join of any graph on $|V_1|$ vertices with an edgeless graph yields an exact graph, provided the edgeless graph has at least the same number of vertices as $|V_1|$. Furthermore, this condition is found to be sufficient for the uniqueness of the optimal solution.

PROPOSITION 3.9. *Let $G(V, E)$ be a graph such that there exists a partition (V_1, V_2) of V , where V_2 is an independent set and every vertex of V_1 is adjacent to every vertex of V_2 . Then, an exact and unique optimal MAXCUT SDP solution exists for G if $|V_1| \leq |V_2|$.*

Proposition 3.9 generalizes known result 9(b) in [Subsection 2.5](#) and proves uniqueness of the optimal solution:

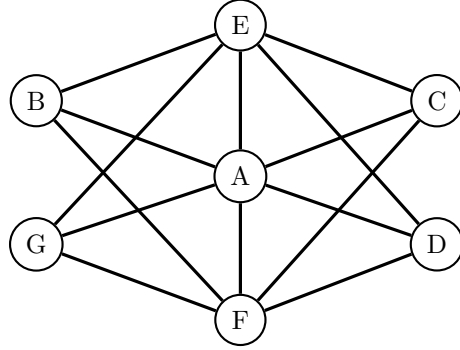


FIG. 3.1. Complete k -partite graph $K(1, 2, 4)$, with $\{A\}$, $\{E, F\}$ and $\{B, C, D, G\}$ as independent sets, is exact with unique solution.

COROLLARY 3.10. *Let $m_1 \leq m_2 \leq \dots \leq m_k \in \mathbb{Z}^+$ and $m_k \geq \sum_{j \neq k} m_j$. Then the complete graph K_n is exact with unique solution if the weights on each edge (i, j) is given by the corresponding pair $\lambda m_i m_j$, where $\lambda \geq 0$. Equivalently, there exists exact and unique optimal MAXCUT SDP solution for the complete k -partite graph $K(m_1 \leq m_2 \leq \dots \leq m_k)$.*

Corollary 3.10 establishes the only known condition under which the MAXCUT SDP admits a unique optimal solution for exact complete graphs. Computational evidence strongly suggests that this condition is also necessary for uniqueness in such graphs. In [Section 4](#), we make a few additional observations on the structure of optimal solutions in exact complete graphs.

In the next result, we consider graphs with a structure similar to that described in [Proposition 3.9](#). Specifically, the join of any graph on $|V_1|$ vertices with an edgeless graph on $|V_2|$ vertices is exact and admits a unique solution, provided that the

maximum degree of any vertex $v \in V_1$ does not exceed $|V_2|/2$.

PROPOSITION 3.11. *Let $G(V, E)$ be a graph such that there exists a partition (V_1, V_2) of V , where V_2 is an independent set and every vertex of V_1 is adjacent to every vertex of V_2 . Then an exact and unique optimal MAXCUT SDP solution exists for G if any vertex in V_1 has degree at most $\frac{|V_2|}{2}$ in the subgraph induced by V_1 .*

COROLLARY 3.12. *The following graphs are exact with unique MAXCUT SDP solution if $n \geq 4$:*

- *The graph join of a cycle graph and an edgeless-graph: $C_m + \overline{K_n}$ (m -gonal n -cone graph).*
- *The graph join of a path graph and an edgeless-graph: $P_m + \overline{K_n}$ (m, n fan graph).*

Remark 3.13. It deserves mention that the size of the independent set $|V_2| \geq 4$ is not a necessary condition for these classes of graphs to be exact. For example, the 3-gonal 3-cone graph, the (2,3)-fan graph, and the (3,3)-fan graph are all exact. On the other hand, 4-gonal 3-cone graph and (4,3)-fan graph are not.

We now turn to investigate the conditions for exactness of lexicographic product of graphs. The lexicographic product of two graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ can be explained informally as the graph obtained by associating a copy of G_2 with each vertex of G_1 , and then, for every edge $(u, u') \in E_1$, connecting all vertices in the copy of G_2 associated with u to all vertices in the copy associated with u' , thereby forming a complete bipartite connection between the corresponding vertex sets. We show that the exactness of G_1 is a sufficient condition for the exactness of the lexicographic product $G_1 \circ G_2$, invariant of the structure of G_2 .

THEOREM 3.14. *$G_1(V_1, E_1) \circ G_2(V_2, E_2)$ is exact if G_1 is exact.*

It immediately follows by induction that $G_1 \circ G_2 \circ \dots \circ G_n$ is exact if G_1 is exact. The exactness (or even bipartiteness) of G_2 , however, does not guarantee the exactness of $G_1 \circ G_2$. Consider the triangle graph K_3 as G_1 and the path graph P_3 as G_2 as an example. While G_2 is bipartite, $G_1 \circ G_2$ yields an optimal solution of $z_{\text{MAXCUT}}^* = 20.25$, indicating non-exactness. [Theorem 3.14](#), combined with the properties of the optimal MAXCUT SDP value z_{MAXCUT}^* , enables us to prove the following:

COROLLARY 3.15. *Consider a uniformly weighted exact graph G on n vertices, where each edge has weight w . Construct a split graph $\tilde{G}$ on G with m vertices $v_{i1}, \dots, v_{im}$ for each vertex $v_i \in G$. For each $1 \leq i \leq n$, let $G'_i = (V_i, E_i)$ be a graph on vertex set $V_i = \{v_{ij} \mid 1 \leq j \leq m\}$, such that for every edge $e \in E_i$, its weight satisfies $w(e) \leq \frac{w}{m^2}$. Then the union $H = \tilde{G} \cup G'_1 \cup \dots \cup G'_n$ is an exact graph.*

[Corollary 3.15](#) further implies that if the vertices of an unweighted exact graph are split into groups of m vertices each—while keeping the join edges unweighted—and any additional unweighted edges are introduced among the vertices within each group, the resulting graph remains exact. Moreover, [Corollary 3.15](#) provides an alternative proof of [Proposition 3.4](#), as can be readily verified.

With these tools in hand, we are now ready to analyze Example 1.

3.2. Example 1 continued. Consider the vertex labeling of the graph of Example 1, as shown in [Figure 3.2](#). We use the fact that $\phi(G)$ is preserved under graph splitting. Also, infer that a graph G is exact if and only if any possible splitting $\tilde{G}$

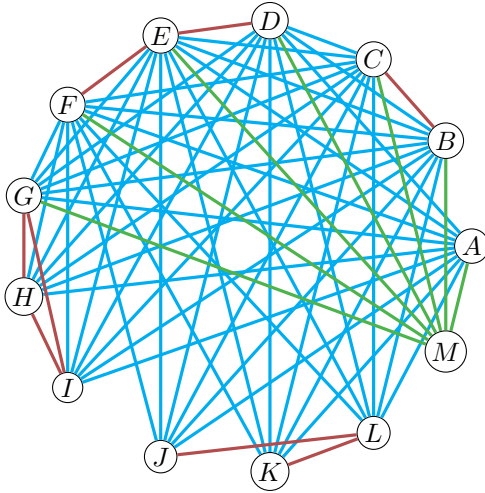


FIG. 3.2. *Labeled Graph $\mathcal{G}$. Edge colors indicate weights: Red (1), Blue (2), and Green (6).*

of G is exact (Lemma 3.1). First, we perform a split on the vertex M into three distinct vertices. Consequently, all edges incident to these newly created vertices now have an edge weight of 2. Within the resulting graph, verify the existence of a spanning subgraph $\mathcal{H}$ with uniform edge weights of 2. This subgraph $\mathcal{H}$ can be derived from an underlying graph $\mathcal{H}'$ of uniform edge weight 18 by obtaining split graph of $\mathcal{H}'$ wherein each vertex is divided into three. Notice that $\mathcal{H}'$ is the k -partite graph $K(1, 1, 3)$. By Corollary 3.10, the graph $\mathcal{H}'$ is exact. Furthermore, the edges in $E(G) \setminus E(\mathcal{H})$ have weights 1, satisfying the relation $1 < \frac{18}{32}$. Hence, by Corollary 3.15, we conclude that $\mathcal{G}$ is exact. Using the construction in the proofs (See Subsections 5.3 and 5.5), one can also verify that the maximum cut of $\mathcal{G}$ is the partition $[(A, B, C, D, E, F), (G, H, I, J, K, L, M)]$ with $z_{\text{MAXCUT}} = 108$.

Remark 3.16. Through Example 1, we wish to highlight that provided a decomposition of a graph G , the factorization can be utilized to sufficiently identify exactness of G . However, we do not propose a general algorithm for obtaining such a decomposition. In fact, existence of such a decomposition/factorization algorithm to efficiently decompose a given graph G , to determine exactness of G , remains an open direction for future research.

4. Rank of Optimal Solutions in Exact Graphs. Examining the rank of optimal solutions in exact graphs raises fundamental questions concerning the semi-definite relaxation. In this context, Mirka and Williamson [12] posed a few points of investigations, and in this section we address two of them. We begin with the following conjecture they proposed:

If a graph admits a unique partition corresponding to its maximum cut and the GW-relaxation on the graph is exact, then the rank-1 optimal solution must also be the unique optimal solution.

We construct the following counterexample to show that the conjecture is not true at least on complete graphs (and equivalently, on complete k -partite graphs). Let $m_1 = 5, m_2 = 3, m_3 = m_4 = 4$. Consider the complete graph K_4 with vertex set indexed by $\{1, 2, 3, 4\}$, and define the weight on each edge w_{ij} as $m_i \times m_j$ for

all $i, j \in \{1, \dots, 4\}$. By exhaustive enumeration, it can be verified that the cut $[(1, 2), (3, 4)]$ is the unique maximum cut, with an optimal value of 64. The GW-relaxation on this graph is exact by 9(a) in [Subsection 2.5](#) (and [Proposition 3.4](#)), since $m_1 + m_2 = m_3 + m_4$. The optimal MAXCUT SDP solution, however, is not unique. Apart from the rank-1 optimal solution corresponding to the maximum cut, the following feasible primal solution Z^* achieves the same objective value of 64:

$$Z^* = \begin{bmatrix} 1 & -\frac{1}{3} & -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{3} & 1 & -\frac{1}{6} & -\frac{1}{6} \\ -\frac{1}{2} & -\frac{1}{6} & 1 & -\frac{1}{4} \\ -\frac{1}{2} & -\frac{1}{6} & -\frac{1}{4} & 1 \end{bmatrix}$$

In the same work, the following question was posed:

Let $G = (V, E)$ be a graph with multiple maximum cuts. Suppose the rank-1 reference solutions corresponding to these cuts are optimal primal solutions of the MAXCUT SDP. Are all higher-rank optimal solutions necessarily convex combinations of these reference solutions?

Following counterexample demonstrates that the answer is negative. Consider again the complete graph K_4 with vertex set $\{1, 2, 3, 4\}$, where the edge weights $w_{ij} = m_i \times m_j$ are defined by $m_1 = m_2 = 2$ and $m_3 = m_4 = 4$. The two maximum cuts of this graph are $(\{1, 3\}, \{2, 4\})$ and $(\{1, 4\}, \{2, 3\})$, both achieving an optimal value of 36. These are the only maximum cuts, which can be verified by enumeration. This graph too, is exact by 9(a) in [Subsection 2.5](#) and [Proposition 3.4](#). Let X^* and Y^* denote the rank-1 optimal solutions associated with the two maximum cuts. Consider the following matrix Z^* , which is also feasible for the primal SDP and attains the same objective value of 36:

$$Z^* = \begin{bmatrix} 1 & -\frac{1}{5} & -\frac{1}{5} & -\frac{1}{5} \\ -\frac{1}{5} & 1 & -\frac{1}{5} & -\frac{1}{5} \\ -\frac{1}{5} & -\frac{1}{5} & 1 & -\frac{4}{5} \\ -\frac{1}{5} & -\frac{1}{5} & -\frac{4}{5} & 1 \end{bmatrix}$$

While Z^* is optimal, it cannot be expressed as a convex combination of X^* and Y^* . This follows from observing that both $X_{12}^* = Y_{12}^* = -1$, whereas $Z_{12}^* = -\frac{1}{5}$. From this observation we can conclude that the higher-rank optimal solutions are not necessarily convex combinations of the rank-1 reference solutions for an exact graph.

5. Proofs. To prove exactness for the graph classes considered in this manuscript, we construct primal and dual solutions that achieve the same objective value and verify their feasibility. From the primal optimal solution, a maximum cut can then be explicitly extracted for these graphs.

5.1. Proof of [Proposition 3.4](#).

Proof. Observe that the graphs G_K, G_A, G_B have no edges in common. The Laplacian matrix of the union graph G' is therefore, simply the sum of the individual Laplacians: $\mathcal{L}_{G'} = \mathcal{L}_{G_K} + \mathcal{L}_{G_A} + \mathcal{L}_{G_B}$. Since $K(n, n)$ is an n -regular bipartite graph, $\lambda_{\max}(\mathcal{L}_{G_K}) = 2n$ and it follows that $\lambda_{\max}(\mathcal{L}_{G'}) \geq \lambda_{\max}(\mathcal{L}_{G_K}) = 2n$. Using property

of the Laplacian, we conclude that $\lambda_{\max}(\mathcal{L}_{G'}) = 2n$. Now, take $\mathbf{x}^\top = \mathbf{1}_A - \mathbf{1}_B$. Then z_{MAXCUT} at $\mathbf{X} = \mathbf{x}\mathbf{x}^\top$ is n^2 . We get the same value for MAXCUT DUAL by taking $Y = \frac{n}{2} \times I$. Furthermore, the dual slack matrix $S = Y - \frac{1}{4}\mathcal{L}_{G'}$ is positive semidefinite, since: $\lambda_{\min}(Y) - \lambda_{\max}(\frac{1}{4}\mathcal{L}_{G'}) = 0$. Thus, the dual constraint is satisfied, and by strong duality $X = \mathbf{x}\mathbf{x}^\top$ is an optimal solution. $\square$

5.2. Proof of Proposition 3.6.

Proof. Let $|V_A| = m$ and $|V_B| = n$. Since G_K, G_A, G_B have no edges in common, we have $\lambda_{\max}(\mathcal{L}_{G'}) = m + n$.

Let $\mathbf{x} = [\mathbf{x}_A, \mathbf{x}_B]^\top$, where $\mathbf{x}_A, \mathbf{x}_B$ are indicator vectors for optimal maxcut solution for G_A and G_B , respectively. This gives $\mathbf{x}^\top \mathcal{L}_{G'} \mathbf{x} = \mathbf{x}^\top \mathcal{L}_{G_K} \mathbf{x} + \mathbf{x}_A^\top \mathcal{L}_{G_A} \mathbf{x}_A + \mathbf{x}_B^\top \mathcal{L}_{G_B} \mathbf{x}_B = \frac{m \times n}{2} + \frac{m^2}{2} + \frac{n^2}{2} = \frac{(m+n)^2}{2}$. For constructing a dual with same objective value, take $\hat{Y} = \frac{m+n}{4} \times \mathbf{I}_{m+n}$. This dual can be verified to be feasible because $\lambda_{\min}(\hat{Y}) - \frac{1}{4}\lambda_{\max}(\mathcal{L}_{G'}) = 0$. $\square$

5.3. Proof of Proposition 3.9. The following lemma is a refinement of the well known Sylvester's inequality for optimal primal-dual pair matrices of GW-relaxation. We will use it to establish uniqueness of the optimal solution in Proposition 3.9 and Proposition 3.11.

LEMMA 5.1. *For any optimal primal (X^*) for MAXCUT SDP and corresponding optimal dual (S^*) for MAXCUT DUAL SDP, the following inequality holds:*

$$\text{rank}(X^*) + \text{rank}(S^*) \leq n - |\mathcal{N}(X^*) \cap \mathcal{N}(S^*)|.$$

Proof. Let $\{a_1, a_2, \dots, a_r\}$ be a basis of $\mathcal{N}(S^*)$. Since $\mathcal{N}(S^*) \subseteq \mathcal{N}(X^*S^*)$, this basis can be extended to form a basis of $\mathcal{N}(X^*S^*)$. Since (X^*, S^*) is an optimal pair, we have $|\mathcal{N}(X^*S^*)| = n$ by duality. Let $\{a_1, a_2, \dots, a_r, a_{r+1}, \dots, a_n\}$ be such a basis of $\mathcal{N}(X^*S^*)$. Observe that $\{S^*(a_{r+1}), \dots, S^*(a_n)\}$ is a set of linearly independent vectors, and $a \in \mathcal{N}(X^*S^*) \implies S^*(a) \in \mathcal{N}(X^*)$. It follows that $n = |\mathcal{N}(X^*S^*)| \leq |\mathcal{N}(X^*)| + |\mathcal{N}(S^*)| - |\mathcal{N}(X^* \cap S^*)|$. The desired inequality follows from the rank-nullity theorem. $\square$

We exploit the block matrix structure of the dual to establish its feasibility for the following proof of Proposition 3.9 (and later for the proof of Proposition 3.11). To prove uniqueness of the solution, we argue that rank > 1 primal solution cannot exist for the respective graphs. Then the rank-1 optimal solution must be unique since the existence of multiple optimal rank-1 matrices implies the existence of higher rank optimal matrices (See [12, Theorem 5]).

Proof. Let $|V_1| = m, |V_2| = k$. WLOG assume $V_1 = \{1, 2, \dots, m\}$ and $V_2 = \{m+1, m+2, \dots, m+k\}$.

To see exactness, consider Y^* such that,

$$Y_{ij}^* = \begin{cases} \frac{k}{2} & \text{if } 1 \leq i \leq m, i = j \\ \frac{m}{2} & \text{if } m+1 \leq i \leq m+k, i = j \\ 0 & \text{if } i \neq j \end{cases}$$

Taking $x_i = +1$ for $i \in V_1$ and $x_j = -1$ for $j \in V_2$ we have $z_{\text{MAXCUTDUAL}} = z_{\text{MAXCUT}} = m \times k$.

For Y^* , the dual feasibility condition can be expressed as a condition on the block

matrix $S^* = Y^* - \frac{1}{4}\mathcal{L}_G = \begin{bmatrix} A & B \\ B^\top & D \end{bmatrix} \succeq \mathbf{0}$. Verify that D is diagonal since the bottom $k \times k$ submatrix corresponds to an independent set. B is $\frac{1}{4}\mathbf{1}_{m \times k}$ by virtue of G being a graph join. Therefore, $D \succ 0$ and $BD^{-1}B^\top$ exists with $(BD^{-1}B^\top)_{ij} = \frac{k}{4 \times m}$ for all i, j . We can now verify that $A - BD^{-1}B^\top$ is diagonally dominant with non-negative diagonal entries and hence positive semidefinite.

For the uniqueness argument, we divide the proof into two cases: $m < k$ and $m = k$. For $m < k$, we use the fact that for any primal-dual optimal solution pair (X^*, S^*) , $\text{rank}(X^*) + \text{rank}(S^*) \leq m + k$, which follows directly from [Lemma 5.1](#). We aim to prove that $\text{rank}(S^*) \geq m + k - 1$.

The submatrix A , formed by restricting S^* to its first m rows and m columns, is full rank whenever $m < k$, since

$$\lambda_{\min} \left[\left(Y^* - \frac{1}{4}\mathcal{L}_G \right) \upharpoonright_{m \times m} \right] \geq \lambda_{\min}(Y^* \upharpoonright_{m \times m}) - \frac{1}{4}\lambda_{\max}(\mathcal{L}_G \upharpoonright_{m \times m}) = \frac{k}{2} - \frac{m+k}{4} > 0.$$

It follows,

$$\begin{aligned} \text{rank}(S^*) &= \text{rank} \left(\begin{bmatrix} A & B \\ B^\top & D \end{bmatrix} \right) \\ &= \text{rank}(D) + \text{rank}(A - BD^{-1}B^\top) \\ &\geq \text{rank}(D) + \text{rank}(A) - \text{rank}(BD^{-1}B^\top) \\ &= k + m - 1. \end{aligned}$$

For $m = k$, any vector $v \in \mathcal{N}(A)$ can be naturally extended to a vector $v^* \in \mathbb{R}^{2m}$ with zeros in the last m coordinates such that $v^* \in \mathcal{N}(X^*) \cap \mathcal{N}(S^*)$. This is apparent from the fact that X^* restricted to leading $m \times m$ principal submatrix is just ones matrix $\mathbf{1}_{m \times m}$ and $\mathbf{1}$ is an eigenvector of A . Let $|\mathcal{N}(X^*) \cap \mathcal{N}(S^*)| = r$. It follows that $\text{rank}(X^*) + \text{rank}(S^*) \leq 2m - r$. Moreover from the earlier discussion, $\text{rank}(S^*) \geq \text{rank}(D) + \text{rank}(A) - \text{rank}(BD^{-1}B^\top) \geq m + (m - r) - 1$. Applying [Lemma 5.1](#), any optimal X^* must be a rank-1 matrix. Multiple optimal rank-1 matrices will imply existence of higher (> 1) ranked optimal matrices by taking any convex combination of the optimal matrices. Hence the optimal X^* must also be unique. $\square$

5.4. Proof of [Proposition 3.11](#).

Proof. Let $G = G_m + \overline{K_n}$, i.e., $|V_1| = m$, $|V_2| = n$. Let $d = \lfloor \frac{n}{2} \rfloor$. Without loss of generality, assume $\{1, 2, \dots, m\} = V_1$ and $\{m+1, m+2, \dots, m+n\} = V_2$. We focus on the case $m > n$, as for $m \leq n$, the result follows from [Proposition 3.9](#).

Similar to the proof of [Proposition 3.9](#), we construct the primal as $x_i = +1$ for $i \in V_1$ and $x_j = -1$ for $j \in V_2$. For the dual solution, consider Y^* as a diagonal matrix, where the first m diagonal entries are $\frac{n}{2}$, and the remaining diagonal entries are $\frac{m}{2}$.

Clearly, we have $z_{\text{MAXCUTDUAL}} = z_{\text{MAXCUT}} = m \times n$. We take $S^* = \begin{bmatrix} A & B \\ B^\top & D \end{bmatrix}$, where $A = Y^* \upharpoonright_{m \times m} - \frac{1}{4}(\mathcal{L}_{G_m} + n\mathbf{I}_m)$, $D = \frac{m}{4}\mathbf{I}_n$, $B = \frac{1}{4}\mathbf{1}_{m \times n}$.

Now, for any given graph G_m with maximum vertex degree d ,

$$\lambda_{\min}(A) \geq \lambda_{\min}(Y^* \upharpoonright_{m \times m}) - \lambda_{\max} \left[\frac{1}{4}(\mathcal{L}_{G_m} + n\mathbf{I}_m) \right] = \frac{n - \lambda_{\max}(\mathcal{L}_{G_m})}{4} \geq 0,$$

where the final inequality follows since $\lambda_{\max}(\mathcal{L}_{G_m}) \geq 2d$. This implies that $A \succeq 0$.

We now analyze the Schur complement of S^* with respect to the block A . Observe that $D - B^\top A^+ B = \frac{m}{4} \mathbf{I}_n - \frac{1}{16} (\mathbf{1}_m^\top A^+ \mathbf{1}_m) \mathbf{1}_{n \times n}$. Since $\mathbf{1}_m$ is an eigenvector of A corresponding to the eigenvalue $\frac{n}{4} > 0$, we have $\mathbf{1}_m^\top A^+ \mathbf{1}_m = \frac{4m}{n}$. Substituting into the expression for the Schur complement yields $D - B^\top A^+ B = \frac{m}{4} I_n - \frac{m}{4n} \mathbf{1}_{n \times n}$. This is the Laplacian matrix of a complete graph on n vertices with positive and uniform edge weights $\frac{m}{4n}$. Moreover, since $\mathbf{1}_m \notin \mathcal{N}(A)$, it follows that $\text{range}(B) \subseteq \text{range}(A)$. We conclude that $D - B^\top A^+ B \succeq 0$, which establishes the positive semidefiniteness of the dual slack matrix S^* .

To establish uniqueness, note that, any vector $v \in \mathcal{N}(A)$ can be extended to $v^* \in \mathcal{N}(X^*) \cap \mathcal{N}(S^*)$. Applying [Lemma 5.1](#), the result thus follows by the same reasoning used in the uniqueness part of the proof of [Proposition 3.9](#). $\square$

5.5. Proof of [Theorem 3.14](#). We use the following lemmas to prove [Theorem 3.14](#):

LEMMA 5.2. *The Laplacian of $G_1 \circ G_2$ is given by:*

$$\mathcal{L}_{G_1 \circ G_2} = \mathbf{I}_m \otimes \mathbb{D}_{G_2} + \mathbb{D}_{G_1} \otimes n\mathbf{I}_n + \mathbf{I}_m \otimes (\mathcal{L}_{G_2} - \mathbb{D}_{G_2}) + (\mathcal{L}_{G_1} - \mathbb{D}_{G_1}) \otimes \mathbf{1}_{n \times n},$$

where $\mathbb{D}_G$ is the degree graph of graph G .

Proof. By construction of the lexicographic product, the adjacency matrix $\mathbb{A}$ of $G_1 \circ G_2$ is $\mathbb{A}(G_1 \circ G_2) = \mathbf{I}_m \otimes \mathbb{A}(G_2) + \mathbb{A}(G_1) \otimes \mathbf{1}_{n \times n}$ and the degree matrix of $G_1 \circ G_2$ is $\mathbb{D}(G_1 \circ G_2) = \mathbf{I}_m \otimes \mathbb{D}(G_2) + \mathbb{D}(G_1) \otimes n\mathbf{I}_n$, where m, n are cardinalities of vertex sets of G_1 and G_2 , respectively. Proof follows from the identity $\mathcal{L} = \mathbb{D} - \mathbb{A}$. (See [\[2\]](#) for details on spectra of lexicographic product). $\square$

LEMMA 5.3. *If $(\bar{X}, \bar{S})$ is a primal-dual optimal pair for any exact graph $G(V, E)$, then $Y_{ii} = \bar{S}_{ii} + \frac{1}{4} \mathcal{L}_{G_{ii}} = \frac{\delta_i}{2}$, where δ_i is the cut (induced by $\bar{X}$) incident on vertex i . Furthermore, $\delta_i \geq \frac{d_i}{2}$ for every node i where d_i is the degree of node i .*

Proof. Verify that $\text{tr}(Y) = \text{MAXCUT}(G_1)$. Since G is exact, $\bar{X} = xx^\top$ for some vector x which induces the max-cut. Since $\bar{S} \succeq 0$ and optimal, $x^\top \bar{S} x = 0 \implies \bar{S} x = 0 \implies \bar{S}_{ii} x = 0$ for every row i . This implies that for all $i \in V$, $y_{ii} - \frac{1}{4}(\delta_i + \delta_i^c) - \frac{1}{4}\delta_i + \frac{1}{4}\delta_i^c = 0 \implies y_{ii} = \frac{\delta_i}{2}$. For the second part, $S = Y - \frac{1}{4}\mathcal{L}_G \succeq 0 \implies S_{ii} \geq 0 \implies \frac{\delta_i}{2} - \frac{d_i}{4} \geq 0 \implies \delta_i \geq \frac{d_i}{2}$. $\square$

[Lemmas 5.2](#) and [5.3](#) generalize for graphs with non-negative edge weights, where d_i is replaced by total weight incident on node i and the degree matrix $\mathbb{D}_G$ is replaced by weight matrix $\mathbb{W}$. We are now ready to prove [Theorem 3.14](#).

Proof. Let $|V_1| = m$ and $|V_2| = n$. Identify each vertex in $G_1 \circ G_2$ as x_{ij} where $i \in V_1$ and $j \in V_2$. Assume that G_1 is connected, otherwise we can extend the proof for each connected component. Let $(\bar{X}, \bar{S})$ correspond to the primal and dual optimal solution of exact graph G_1 . Let (V_A, V_B) be the partition corresponding to $\bar{X}$ for value $\text{MAXCUT}(G_1)$. We claim that, $\hat{X} = \hat{x}\hat{x}^\top$ is optimal for $G_1 \circ G_2$ if $\hat{x}_{ij} = +1$ for $i \in V_A$ and $\hat{x}_{ij} = -1$ for $i \in V_B$. Verify that $z_{\text{MAXCUT}} = n^2 \text{MAXCUT}(G_1)$ for $\hat{X}$.

Let $\bar{Y} = \bar{S} + \frac{1}{4}\mathcal{L}_{G_1}$. Take $\hat{Y} = \bar{Y} \otimes n\mathbf{I}_n$. We claim that $\hat{S} = \hat{Y} - \frac{1}{4}\mathcal{L}_{G_1 \circ G_2}$ is the dual optimal solution. Verify that $\text{tr}(\hat{Y}) = n^2 \text{tr}(\bar{Y}) = n^2 \text{MAXCUT}(G_1)$. We reformulate

$\hat{S}$ using [Lemma 5.2](#).

$$\begin{aligned}
\hat{S} &= \bar{Y} \otimes (\mathcal{L}_{K_n} + \mathbf{1}_{n \times n}) - \frac{1}{4} \left[\mathbb{D}_{G_1} \otimes (\mathcal{L}_{K_n} + \mathbf{1}_{n \times n}) + \mathbf{I}_m \otimes (\mathcal{L}_{G_2} - \mathbb{D}_{G_2}) \right. \\
&\quad \left. + (\mathcal{L}_{G_1} - \mathbb{D}_{G_1}) \otimes \mathbf{1}_{n \times n} + \mathbf{I}_m \otimes \mathbb{D}_{G_2} \right] \\
&= (\bar{Y} - \frac{1}{4} \mathbb{D}_{G_1}) \otimes \mathcal{L}_{K_n} + (\bar{Y} - \frac{1}{4} \mathcal{L}_{G_1}) \otimes \mathbf{1}_{n \times n} - \frac{1}{4} \mathbf{I}_m \otimes (\mathcal{L}_{K_n} - \mathcal{L}_{G_2}^c) \\
&= \underbrace{(\bar{Y} - \frac{1}{4} \mathbb{D}_{G_1} - \frac{1}{4} \mathbf{I}_m) \otimes \mathcal{L}_{K_n}}_{Y_1} + \underbrace{(\bar{Y} - \frac{1}{4} \mathcal{L}_{G_1}) \otimes \mathbf{1}_{n \times n}}_{Y_2} + \underbrace{\frac{1}{4} \mathbf{I}_m \otimes (\mathcal{L}_{G_2}^c)}_{Y_3},
\end{aligned}$$

where the penultimate equation follows as $\mathcal{L}_G + \mathcal{L}_G^c = \mathcal{L}_K$. Since $\bar{Y}$ is optimal for G_1 , $\lambda_{\min}(Y_2) = 0$ and it is clear that $\lambda_{\min}(Y_3) = 0$. Referring to [Lemma 5.3](#), note that for every vertex i in a connected and exact graph G , $Y_{ii} - \frac{1}{4} \mathbb{D}_{G_{ii}} \geq \frac{1}{4}$. It follows that $\lambda_{\min}(Y_1) = 0$. $\square$

5.5.1. Proof of [Corollary 3.15](#).

Proof. Notice that $\tilde{G}$ is a uniformly weighted graph with weights $\frac{w}{m^2}$ and exact with $\phi(\tilde{G}) = \text{MAXCUT}(G)$ (See [Lemma 3.1](#)). Let G_w denote a copy of the graph G in which all edge weights are uniform and equal to $\frac{w}{m^2}$, and let $K_{m,w}$ be a complete graph with the same uniform edge weights. Consider the lexicographic product $G_w \circ K_{m,w}$. By applying [Theorem 3.14](#), we conclude that $G_w \circ K_{m,w}$ is exact with $\phi(G_w \circ K_{m,w}) = m^2 \times \text{MAXCUT}(G_w) = \text{MAXCUT}(G)$. Observe that the graphs $\tilde{G}$, H , and $G_w \circ K_{m,w}$ share the same vertex set and the weights on edges of graph H lies between the corresponding edges of two extremes $\tilde{G}$ and $G_w \circ K_{m,w}$. The desired result then follows from the monotonicity property of ϕ . $\square$

6. Conclusion. To begin with, we showed that the decision problem of recognition of exact unweighted graphs is NP -complete. We then identified the following classes of unweighted graphs for which the GW-relaxation of the Max-Cut problem is exact:

1. Graphs with balanced complete bipartite spanning subgraph.
2. Graph join of two graphs G_A and G_B , each with balanced complete bipartite spanning subgraph.
3. The lexicographic product of G_1 and G_2 where G_1 is an exact graph.

Let $G(V, E)$ be a graph such that there exists a partition (V_1, V_2) of V , where V_2 is an independent set and every vertex of V_1 is adjacent to every vertex of V_2 . Then, G was shown to be exact with unique solution if:

4. The size of the independent set V_2 is atleast the size of V_1 .
5. Any vertex in V_1 has degree at most $\frac{|V_2|}{2}$ in the subgraph induced by V_1 .

Furthermore, we presented special cases of these graph classes in [Section 3](#) and provided explicit constructions of their maximum cut in [Section 5](#). Prior to this work, the only exact graph families known to exhibit uniqueness of the optimal solution were appropriately weighted cycle graphs [\[7\]](#) and bipartite graphs with non-negative edge weights [\[12\]](#). We expanded this list by identifying graph classes (4) and (5). Using [Example 1](#), we demonstrated that our findings lay the groundwork for identifying the exactness of more intricate graphs. We also address two open problems

posed by Mirka-Williamson [12]. First, we provide a counterexample to their conjecture that the exactness of GW-relaxation combined with uniqueness of the maximum cut implies uniqueness of the corresponding rank-1 optimal solution. Second, we answer whether all higher-rank optimal solutions of an exact relaxation necessarily lie in the convex hull of rank-1 solutions associated with the maximum cuts in the negative. Both problems are addressed via explicit counterexamples on weighted complete graphs. A complete characterization of when the optimal solution to a given exact GW-relaxation is unique, however, remains an open problem. As most of our results concern unweighted (or equivalently, uniformly weighted) graphs, it would be worthwhile to investigate further the conditions under which GW-relaxation yield exact solutions for non-uniformly weighted instances. We believe that advancing these lines of inquiry would enhance the applicability of the theoretical framework and provide deeper insights into the structural properties that underlie exactness.

REFERENCES

- [1] F. BARAHONA, M. GRÖTSCHEL, M. JÜNGER, AND G. REINELT, *An application of combinatorial optimization to statistical physics and circuit layout design*, Operations Research, 36 (1988), pp. 493–513.
- [2] S. BARIK, R. B. BAPAT, AND S. PATI, *On the laplacian spectra of product graphs*, Applicable Analysis and Discrete Mathematics, (2015), pp. 39–58.
- [3] C. DELORME AND S. POLJAK, *Combinatorial properties and the complexity of a max-cut approximation*, European Journal of Combinatorics, 14 (1993), pp. 313–333.
- [4] C. DELORME AND S. POLJAK, *Laplacian eigenvalues and the maximum cut problem*, Mathematical Programming, 62 (1993), pp. 557–574.
- [5] C. DELORME AND S. POLJAK, *The performance of an eigenvalue bound on the max-cut problem in some classes of graphs*, Discrete Mathematics, 111 (1993), pp. 145–156.
- [6] M. X. GOEMANS AND D. P. WILLIAMSON, *Improved approximation algorithms for maximum cut and satisfiability problems using semidefinite programming*, Journal of the ACM (JACM), 42 (1995), pp. 1115–1145.
- [7] D. HONG, H. LEE, AND A. WEI, *Optimal solutions and ranks in the max-cut sdp*, arXiv preprint arXiv:2109.02238, (2021).
- [8] H. KARLOFF, *How good is the goemans-williamson max cut algorithm?*, in Proceedings of the twenty-eighth annual ACM symposium on Theory of Computing, 1996, pp. 427–434.
- [9] R. M. KARP, *Reducibility among combinatorial problems*, in 50 Years of Integer Programming 1958-2008: from the Early Years to the State-of-the-Art, Springer, 2009, pp. 219–241.
- [10] S. KHOT, G. KINDLER, E. MOSSEL, AND R. O'DONNELL, *Optimal inapproximability results for max-cut and other 2-variable csps?*, SIAM Journal on Computing, 37 (2007), pp. 319–357.
- [11] M. LAURENT AND S. POLJAK, *On a positive semidefinite relaxation of the cut polytope*, Linear Algebra and its Applications, 223 (1995), pp. 439–461.
- [12] R. MIRKA AND D. P. WILLIAMSON, *Max cut and semidefinite rank*, Operations Research Letters, 53 (2024), p. 107067.
- [13] B. MOHAR AND S. POLJAK, *Eigenvalues and the max-cut problem*, Czechoslovak Mathematical Journal, 40 (1990), pp. 343–352.
- [14] S. POLJAK AND F. RENDL, *Nonpolyhedral relaxations of graph-bisection problems*, SIAM Journal on Optimization, 5 (1995), pp. 467–487.
- [15] S. SOJODI AND J. LAVAEI, *Exactness of semidefinite relaxations for nonlinear optimization problems with underlying graph structure*, SIAM Journal on Optimization, 24 (2014), pp. 1746–1778.
- [16] L. VANDENBERGHE AND S. BOYD, *Semidefinite programming*, SIAM review, 38 (1996), pp. 49–95.
- [17] J. WANG, T. JEBARA, AND S.-F. CHANG, *Semi-supervised learning using greedy max-cut*, The Journal of Machine Learning Research, 14 (2013), pp. 771–800.